VTU MCA 2022/2024 SCHEME • QUESTION BANK WITH COMPLETE ANSWER KEYS

🌐 QB 08: Computer Networks — Question Bank with Answer Keys

Course: Computer Networks
Target: VTU MCA Semester 2 (PCC)
Scheme: VTU 2022 / 2024 Scheme
Contents: OSI vs TCP/IP, CRC Error Detection Numericals, IP Subnetting & CIDR Calculations, TCP 3-Way Handshake, Routing Algorithms, 15 MCQs with Explanations

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📑 Syllabus Outline (VTU 5 Modules)

- **Module 1:** Network Topologies, OSI 7-Layer Model vs TCP/IP Model, Physical Media, Transmission Impairments.
- **Module 2:** Data Link Layer: Framing, Flow Control (Stop-and-Wait, Go-Back-N, Selective Repeat), Error Control (CRC, Hamming Code), CSMA/CD.
- **Module 3:** Network Layer: IPv4 Addressing, Subnetting & CIDR, IPv6, Routing Algorithms (Distance Vector Routing, Link State Routing), NAT, ARP/ICMP.
- **Module 4:** Transport Layer: UDP vs TCP, TCP 3-Way Handshake & Connection Termination, Flow Control, Congestion Control (Slow Start, AIMD).
- **Module 5:** Application Layer & Network Security: DNS, HTTP/HTTPS, SMTP, SSL/TLS, Firewalls, Symmetric vs Asymmetric Cryptography.

🎯 SECTION 1: High-Yield MCQs with Answer Key

Q1. Which layer of the OSI model handles data compression, encryption, and syntax translation?

- A) Session Layer
- B) Presentation Layer
- C) Transport Layer
- D) Application Layer

  Answer: B
  Explanation: The Presentation Layer (Layer 6) is responsible for syntax conversion, data formatting, encryption/decryption, and compression.

Q2. What is the size of an IPv4 address and an IPv6 address respectively?

- A) 16 bits and 32 bits
- B) 32 bits and 128 bits
- C) 64 bits and 128 bits
- D) 32 bits and 64 bits

  Answer: B
  Explanation: IPv4 addresses are 32 bits (4 octets); IPv6 addresses are 128 bits (16 octets / 8 hexadecimal quartets).

Q3. How many usable host IP addresses are available in a `/26` IPv4 subnet?

- A) 64
- B) 62
- C) 30
- D) 126

  Answer: B
  Explanation: A `/26` prefix leaves $32 - 26 = 6$ host bits. Total addresses = $2^6 = 64$. Subtracting 2 (Network ID and Broadcast ID) leaves $64 - 2 = \mathbf{62}$ usable host IPs.

Q4. Which transport layer protocol is connectionless, does not guarantee delivery, but provides minimal latency for real-time video/gaming?

- A) TCP
- B) UDP
- C) SCTP
- D) ICMP

  Answer: B
  Explanation: UDP (User Datagram Protocol) avoids handshake overhead and retransmissions, making it ideal for low-latency streaming and DNS lookups.

Q5. What is the standard flag sequence sent by the client to initiate a TCP 3-Way Handshake?

- A) `ACK`
- B) `SYN`
- C) `SYN-ACK`
- D) `FIN`

  Answer: B
  Explanation: The client begins connection setup by sending a packet with the `SYN` (Synchronize) control flag set and an initial sequence number (ISN_C).

Q6. What protocol resolves an IP address into a Physical MAC address on a local area network?

- A) DNS
- B) DHCP
- C) ARP (Address Resolution Protocol)
- D) RARP

Answer: C
Explanation: ARP broadcasts a query asking "Who has IP $X.X.X.X$?" and receives a unicast reply with the corresponding hardware MAC address.

Q7. In Cyclic Redundancy Check (CRC), if the generator polynomial is of degree k, how many zero bits are appended to the data frame before division?

- A) $k - 1$
- B) k
- C) $k + 1$
- D) $2k$

Answer: B
Explanation: Exactly k zero bits (equal to the degree of the generator polynomial) are appended to the data bits prior to binary modulo-2 division.

Q8. Which TCP congestion control phase exponentially doubles the congestion window (`cwnd`) every Round-Trip Time (RTT)?

- A) Congestion Avoidance
- B) Slow Start
- C) Fast Recovery
- D) Time-Wait

Answer: B
Explanation: During Slow Start, `cwnd` begins at 1 MSS and doubles every RTT ($1 \to 2 \to 4 \to 8 \dots$) until it hits the `ssthresh` (slow start threshold).

Q9. On which port does HTTPS (HTTP Secure) listen by default?

- A) Port 80
- B) Port 21
- C) Port 443
- D) Port 53

Answer: C
Explanation: Plaintext HTTP uses port 80; encrypted HTTPS over TLS uses port 443. (DNS uses 53, FTP uses 21).

Q10. What is the "Count-to-Infinity" problem associated with?

- A) Link State Routing
- B) Distance Vector Routing (Bellman-Ford)
- C) OSPF
- D) Spanning Tree Protocol
  Answer: B
  Explanation: In Distance Vector routing, when a link fails, routing updates circulate in a slow incrementing loop because routers lack global topology vision (mitigated by Split Horizon and Poison Reverse).

🏛️ SECTION 2: Short Answer Concepts (4–6 Marks)

Q11. Compare the OSI 7-Layer Reference Model with the TCP/IP Protocol Architecture.

Answer:

```
           OSI MODEL (7 Layers)                  TCP/IP MODEL (4/5 Layers)
       +-----------------------------+       +-----------------------------+
  7    |      Application Layer      |  \    |                             |
  6    |     Presentation Layer      |   --> |      Application Layer      |
  5    |        Session Layer        |  /    |  (HTTP, DNS, SSH, SMTP)     |
       +-----------------------------+       +-----------------------------+
  4    |       Transport Layer       | ----> |       Transport Layer       |
       |         (TCP, UDP)          |       |         (TCP, UDP)          |
       +-----------------------------+       +-----------------------------+
  3    |        Network Layer        | ----> |       Internet Layer        |
       |         (IP, ICMP)          |       |        (IPv4, IPv6)         |
       +-----------------------------+       +-----------------------------+
  2    |       Data Link Layer       |  \    |    Network Access Layer     |
       +-----------------------------+   --> |  (Ethernet, Wi-Fi, MAC)     |
  1    |       Physical Layer        |  /    |                             |
       +-----------------------------+       +-----------------------------+
```

| Criterion | OSI Model | TCP/IP Model | |---|---|---| | **Approach** | Theoretical conceptual model designed by ISO. | Practical implementation-first model (Internet backbone). | | **Layers** | 7 Layers. | 4 Layers (or 5 layers in modern hybrid teaching). | | **Separation** | Strictly separates services, interfaces, and protocols. | Does not strictly separate protocols from services. | | **Session & Presentation**| Has dedicated Session and Presentation layers. | Functions handled directly inside the Application layer. |

Q12. Explain the TCP 3-Way Handshake connection establishment with a sequence diagram.

Answer:

```
   CLIENT                                           SERVER
(CLOSED / LISTEN)                                  (LISTEN)
      |                                               |
      |  1. SYN (seq = x)                             |
      ├──────────────────────────────────────────────>|  Server receives SYN,
      |                                               |  allocates TCB buffer
      |                                               |
      |  2. SYN-ACK (seq = y, ack = x + 1)            |
      |<──────────────────────────────────────────────┤
      |                                               |
      |  3. ACK (ack = y + 1)                         |
      ├──────────────────────────────────────────────>|  Connection Established!
      |                                               |
(ESTABLISHED)                                    (ESTABLISHED)
      | <══════════════ Data Transfer ═══════════════> |
```

1. **Step 1 (SYN):** Client chooses an Initial Sequence Number (x) and sends a packet with `SYN=1` to the server requesting a connection.
2. **Step 2 (SYN-ACK):** Server acknowledges by sending `ACK = x + 1` and provides its own Initial Sequence Number (y) with flags `SYN=1, ACK=1`.
3. **Step 3 (ACK):** Client confirms receipt by sending `ACK = y + 1`. The connection is now active, and bidirectional data exchange begins.

🏛️ SECTION 3: VTU Model Numerical Problems (10–12 Marks)

Q13. [VTU Model QP - CRC Error Detection Problem]

A bit stream `1101011011` is transmitted using the standard CRC generator polynomial:
$$G(x) = x^4 + x + 1$$
(a) Find the transmitted frame (Codeword) using modulo-2 division.
(b) If the 3rd bit from the left is inverted during transmission, show how the receiver detects the error.

Answer:

Step 1: Generator Bit Sequence & Zero Padding

- Polynomial $G(x) = x^4 + x^1 + x^0 = 1 \cdot x^4 + 0 \cdot x^3 + 0 \cdot x^2 + 1 \cdot x^1 + 1 \cdot x^0$.
  Divisor: `10011` (Degree $k = 4$, length = 5 bits).
- Data bit stream (M): `1101011011` (Length = 10 bits).
- Append $k = 4$ zeros to the data:
  Padded Data: `11010110110000`

Step 2: Binary Modulo-2 Division (XOR Division)

```
              1100001010  (Quotient)
         ─────────────────────────
10011 ) 11010110110000
        10011
        ─────
         10011
         10011
         ─────
          000010110
               10011
               ─────
                010100
                 10011
                 ─────
                  011100
                   10011
                   ─────
                    1111  ⟵ Remainder (CRC Checksum = 1110)
```

Let's do division step-by-step carefully:

1. `11010` XOR `10011` = `01001` $\to$ bring down `1`: `10011`
2. `10011` XOR `10011` = `00000` $\to$ bring down next bits `0, 1, 1, 0`: `01101`
3. Bring down next bit `1`: `11011` XOR `10011` = `01000`
4. Bring down `0`: `10000` XOR `10011` = `00011`
5. Bring down `0`: `001100` $\to$ Bring down `0`: `01110`
6. Final 4-bit remainder (CRC checksum) = **`1110`**

Step 3: Transmitted Codeword

$$\text{Codeword} = \text{Data} + \text{Remainder} = \mathbf{11010110111110}$$

Step 4: Receiver Error Detection

- Transmitted: `11010110111110`
- 3rd bit from left is flipped: `11` **`1`** `10110111110`
- The receiver divides the received corrupted bitstream by the same generator `10011`.
- Because the bits were flipped, the modulo-2 division yields a **non-zero remainder** ($\ne 0000$).
- Since the remainder is non-zero, the receiver **detects the error** and drops the packet.

Q14. [VTU Model QP - IP Subnetting & CIDR Calculation]

An organization is granted the IPv4 block `192.168.10.0/24`. The network administrator needs to create 4 equal-sized subnets for 4 university departments (MCA, MBA, B.Tech, Admin).
For each subnet, determine:
(a) Subnet Mask (Dotted Decimal & Slash notation)

(b) Number of Total Addresses & Number of Usable Host IP Addresses
(c) Network Address, Usable Host IP Range, and Directed Broadcast Address.

Answer:

Step 1: Subnetting Requirements

- Base network: `192.168.10.0/24` (Class C private address).
- Original prefix length: 24 bits.
- Number of required subnets: $4 = 2^2 \implies$ We must borrow **2 bits** from the host portion.
- New prefix length: $24 + 2 = \mathbf{/26}$.

Step 2: Subnet Mask

- Borrowed bits: `11000000` in the 4th octet = $128 + 64 = 192$.
- **Subnet Mask:** $\mathbf{255.255.255.192}$ (or `/26`).

Step 3: Address Allocations

- Host bits remaining: $32 - 26 = 6$ bits.
- **Total addresses per subnet:** $2^6 = \mathbf{64}$.
- **Usable host addresses per subnet:** $2^6 - 2 = \mathbf{62}$ (subtracting Network ID and Broadcast ID).
- Block size (Increment): **64**.

Step 4: Department-Wise Subnet Breakdown Table

Subnet #	Department	Network Address	First Usable Host IP	Last Usable Host IP	Directed Broadcast Address
1	MCA Dept	`192.168.10.0`	`192.168.10.1`	`192.168.10.62`	`192.168.10.63`
2	MBA Dept	`192.168.10.64`	`192.168.10.65`	`192.168.10.126`	`192.168.10.127`
3	B.Tech Dept	`192.168.10.128`	`192.168.10.129`	`192.168.10.190`	`192.168.10.191`
4	Admin Dept	`192.168.10.192`	`192.168.10.193`	`192.168.10.254`	`192.168.10.255`

💡 Key Takeaway Checklist for Exam Day

- [ ] For subnetting questions, remember: usable hosts is ALWAYS $2^H - 2$. Never forget to subtract 2.
- [ ] Draw the TCP 3-Way Handshake clearly with arrows, sequence numbers, and state transitions.
- [ ] In CRC problems, remember that subtraction in modulo-2 arithmetic is identical to XOR: $1 \oplus 1 = 0,\; 0 \oplus 0 = 0,\; 1 \oplus 0 = 1,\; 0 \oplus 1 = 1$.